

Universal Digital Agents That Learn, Remember, and Act

Research Internship / Talent Program Interview

GUI Agents

Continual Learning

Real Systems

From research questions to working systems and measurable improvement loops

Expected Jun 2028 | Zhejiang University APRIL Lab



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Candidate Snapshot | Research Focus and Impact

Research vision: universal agents that continuously learn, remember, and act in real digital environments

Mobile GUI agents, long-horizon tasks, memory evaluation, and reinforcement-learning post-training

5

First-author works

TMLR / ACL / arXiv

415

Google Scholar citations

Academic impact

122

First-author citations

Core research

632

GitHub stars on first-author projects

Open-source impact

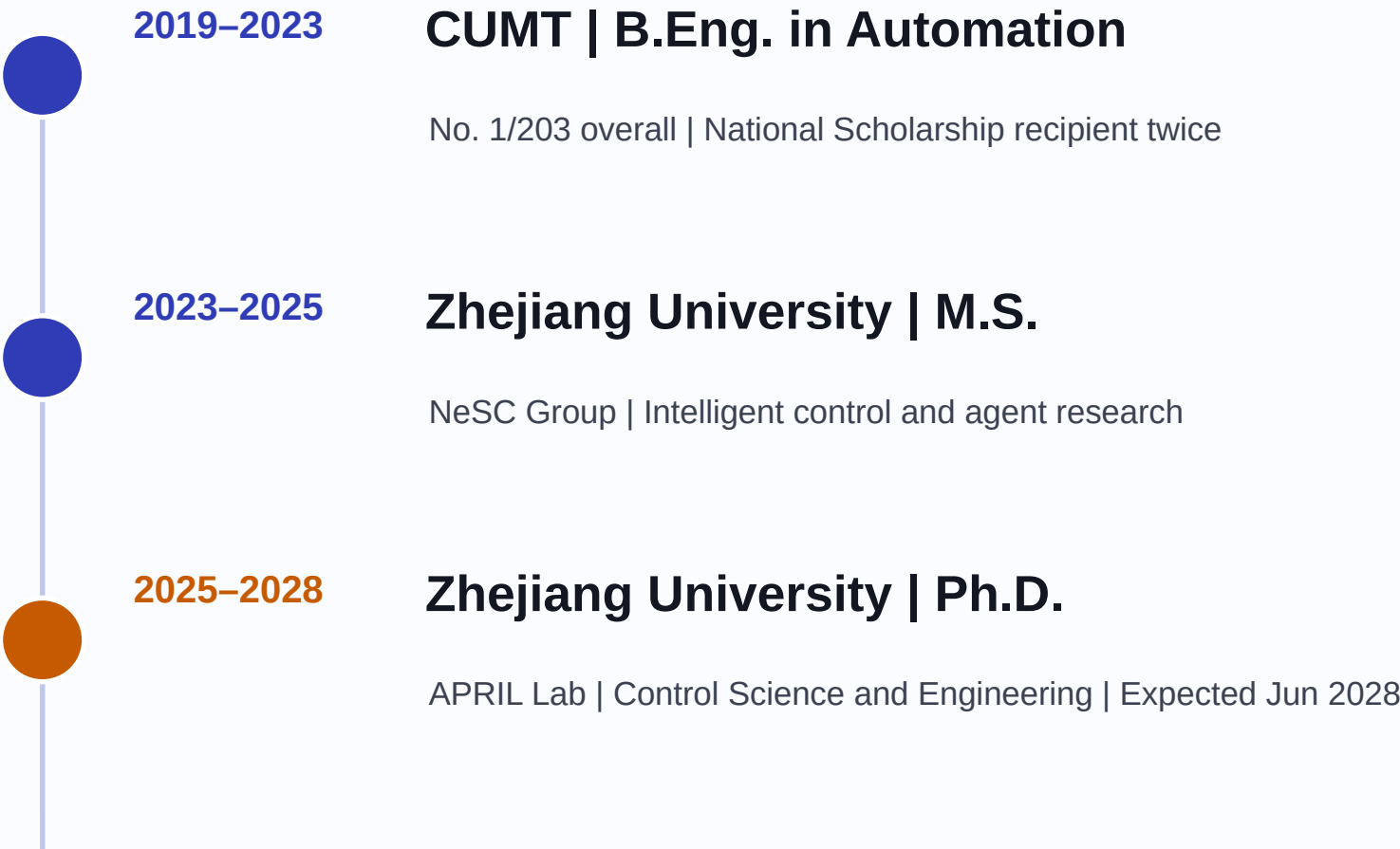
Jun 2028

Expected graduation

Zhejiang University Ph.D.

Source: CV and personal homepage; metrics as of 2026-08-09

Education and Growth



A continuous agent-research trajectory

Source: CV and personal homepage; metrics as of 2026-08-09

Industry Research | From Prototypes to Real Systems

Aug 2024–Aug 2025

vivo AI Lab

- PhoneGPT and demonstration learning
- Online evaluation and memory benchmarking
- Connecting research questions to real phone scenarios

Aug 2025–Jul 2026

Kuaishou

- Agent post-training and reinforcement learning
- Long-horizon context management
- App defect detection and automated evaluation

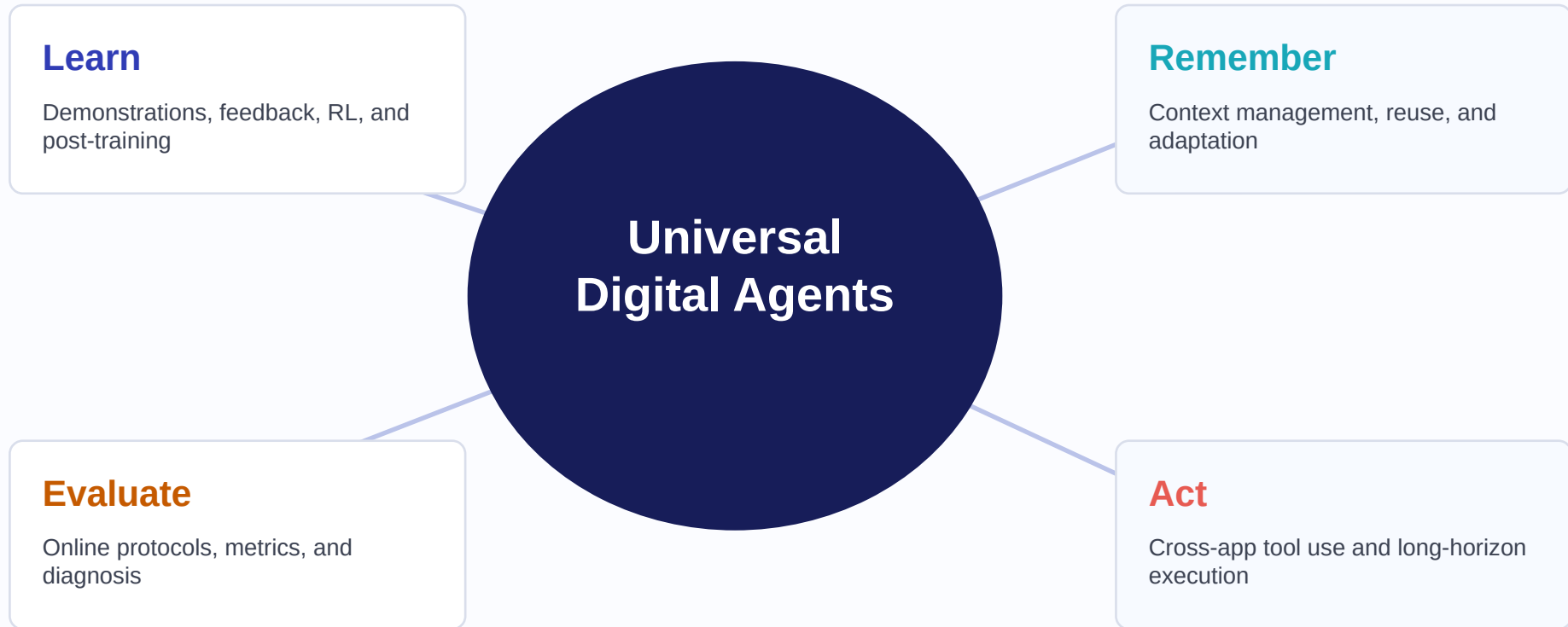
Research | Testable questions

Engineering | Data and eval loops

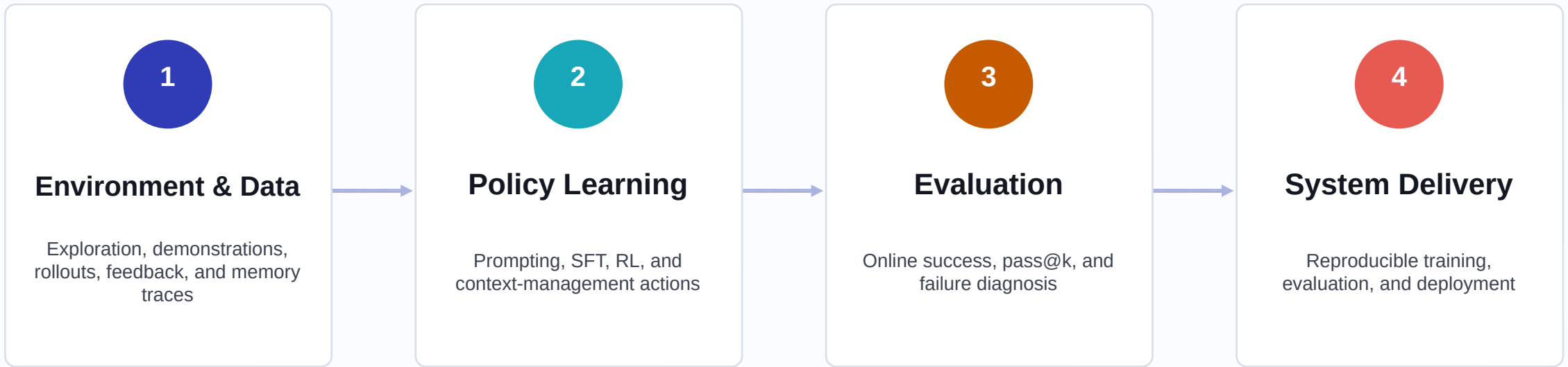
Application | Product workflows

Strength: defining the research question and carrying it through environment, data, training, and evaluation

Research Thesis | Continuous Improvement in Real Environments

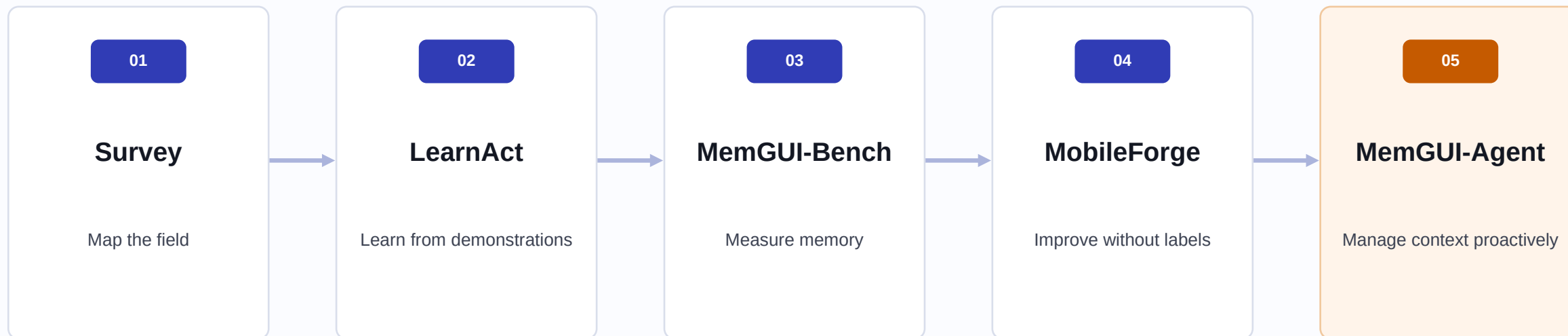


End-to-End Ownership | From Question to System



Find the bottleneck → make it measurable → close the loop → test transfer

Five First-Author Works, One Coherent Research Arc



Field map → demonstration learning → memory evaluation → self-improvement → long-horizon agents

Next: problem, method, evidence, and research value for each paper

LLM-Powered GUI Agents in Phone Automation: Surveying Progress and Prospects

A systematic map of frameworks, models, data, evaluation, and future directions for phone GUI agents

CORE QUESTION A rapidly growing field lacked a shared vocabulary connecting academic progress with practical agent design.

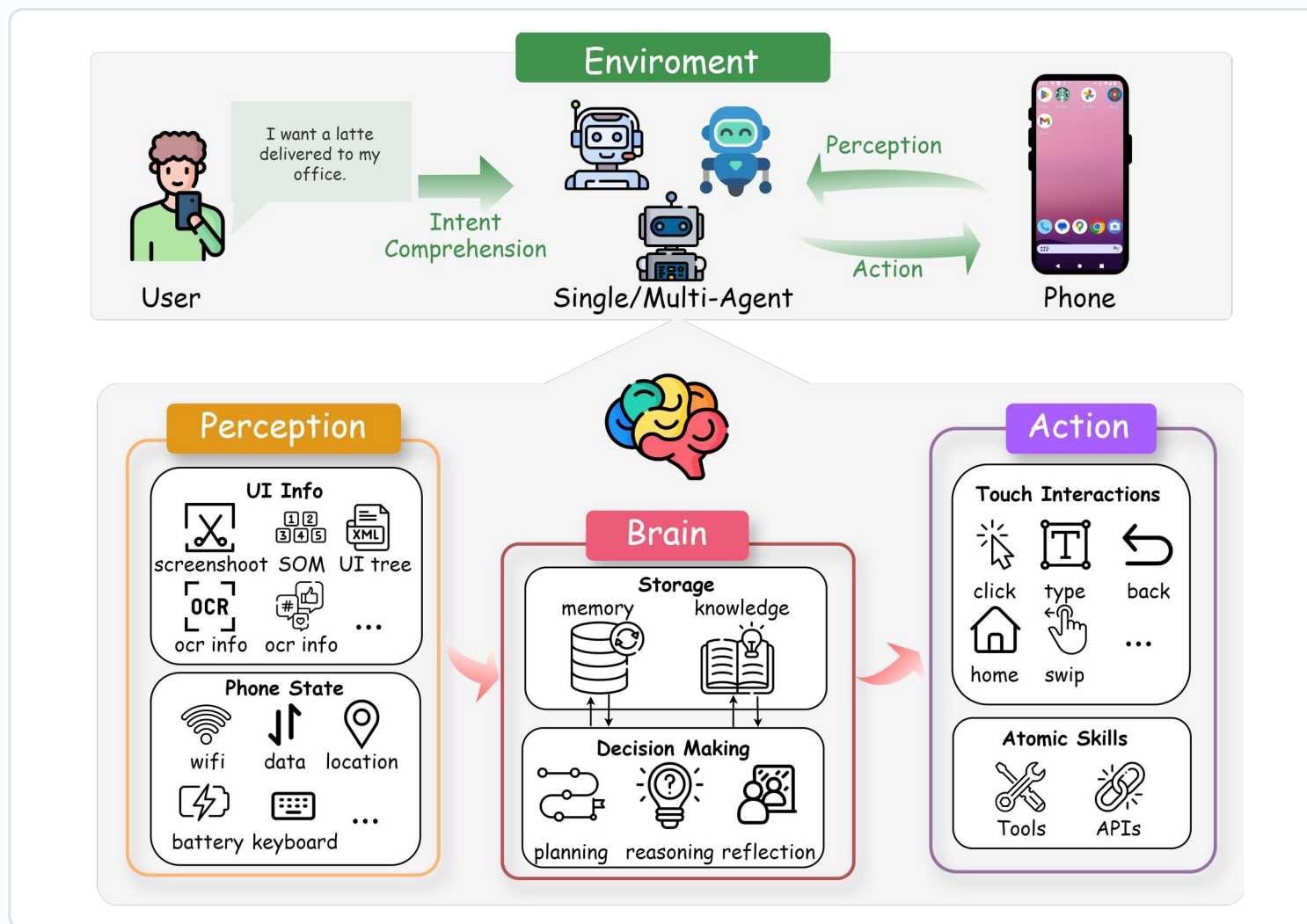
TMLR 2025

Survey & Taxonomy

First Author

arXiv:2504.19838

Unified Framework | From User Intent to GUI Action



A Three-Layer Research Map

- Agent frameworks: single-agent, multi-agent, and plan-then-act
- Modeling: prompt engineering versus training-based methods
- Resources: datasets, benchmarks, metrics, and automated evaluation
- Core loop: perceive the GUI, understand intent, decide, and act

A clear field map reveals the system bottlenecks worth solving

Research Roadmap | Toward Deployable Phone Agents

The shift from static scripts to adaptive agents exposes bottlenecks beyond model capability

Data

Coverage diversity

Long-tail apps and real tasks

On-device

Deployment efficiency

Compute, latency, and energy

Users

Personal adaptation

Preferences, history, and feedback

Safety

Reliability and privacy

Sensitive actions and risk control

RESEARCH VALUE A common vocabulary and evaluation lens that anchors later work on demonstration learning, memory, and continual improvement.

A clear field map reveals the system bottlenecks worth solving

LearnAct: Few-Shot Mobile GUI Agent with a Unified Demonstration Benchmark

Improving long-tail mobile-task generalization from a few visual demonstrations

CORE QUESTION Agents struggle to generalize to long-tail tasks, while the field lacked a unified benchmark for few-shot demonstration learning.

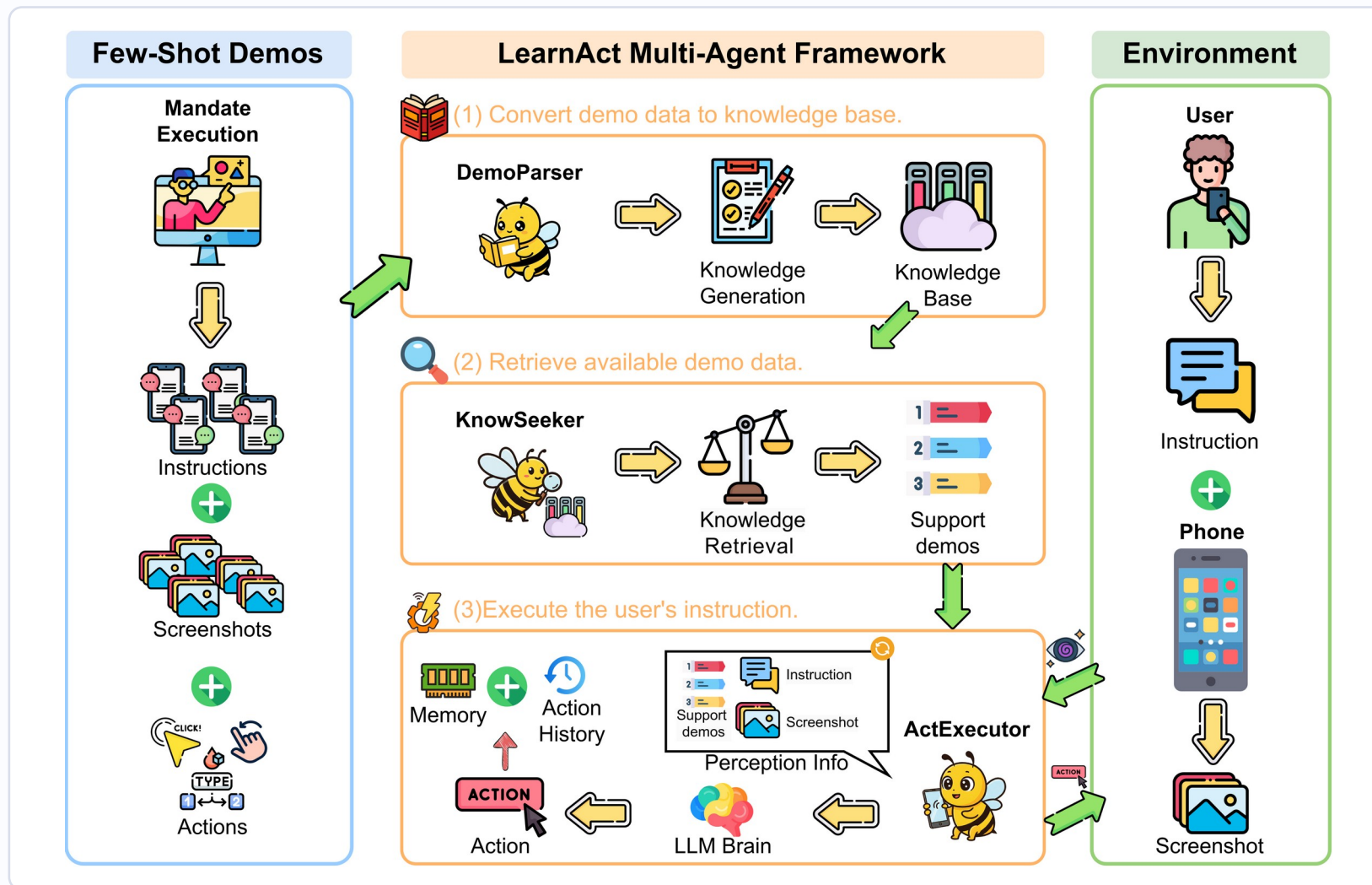
ACL 2026

2,353 Tasks

6 Backbones

arXiv:2504.13805

LearnAct | Turning Visual Demonstrations into Retrievable Knowledge



Three Modules

- DemoParser extracts structured knowledge from demonstrations
- KnowSeeker retrieves support demos by instruction, UI, and action similarity
- ActExecutor combines demos, screenshots, history, and memory for execution

A few high-quality demonstrations can materially improve online generalization

Results | Consistent Gains Across Model Families

LearnGUI: 2,252 offline tasks + 101 online tasks across 73 applications

38.5 → 58.9%

Gemini-2.5-Pro offline

+20.4 percentage points

18.1 → 32.8%

UI-TARS-7B-SFT online

+14.7 pp; +81.2% relative

6

Backbone models

General-purpose and GUI-specialized

KEY FINDING

Demonstration learning benefits both general-purpose and GUI-specialized models, providing a low-cost interface for long-tail adaptation.

A few high-quality demonstrations can materially improve online generalization

MemGUI-Bench: Benchmarking Memory of Mobile GUI Agents in Dynamic Environments

A systematic benchmark for short-term memory, long-term learning, and execution efficiency

CORE QUESTION Existing benchmarks contain only 5.2–11.8% memory-related tasks and cannot evaluate cross-session learning.

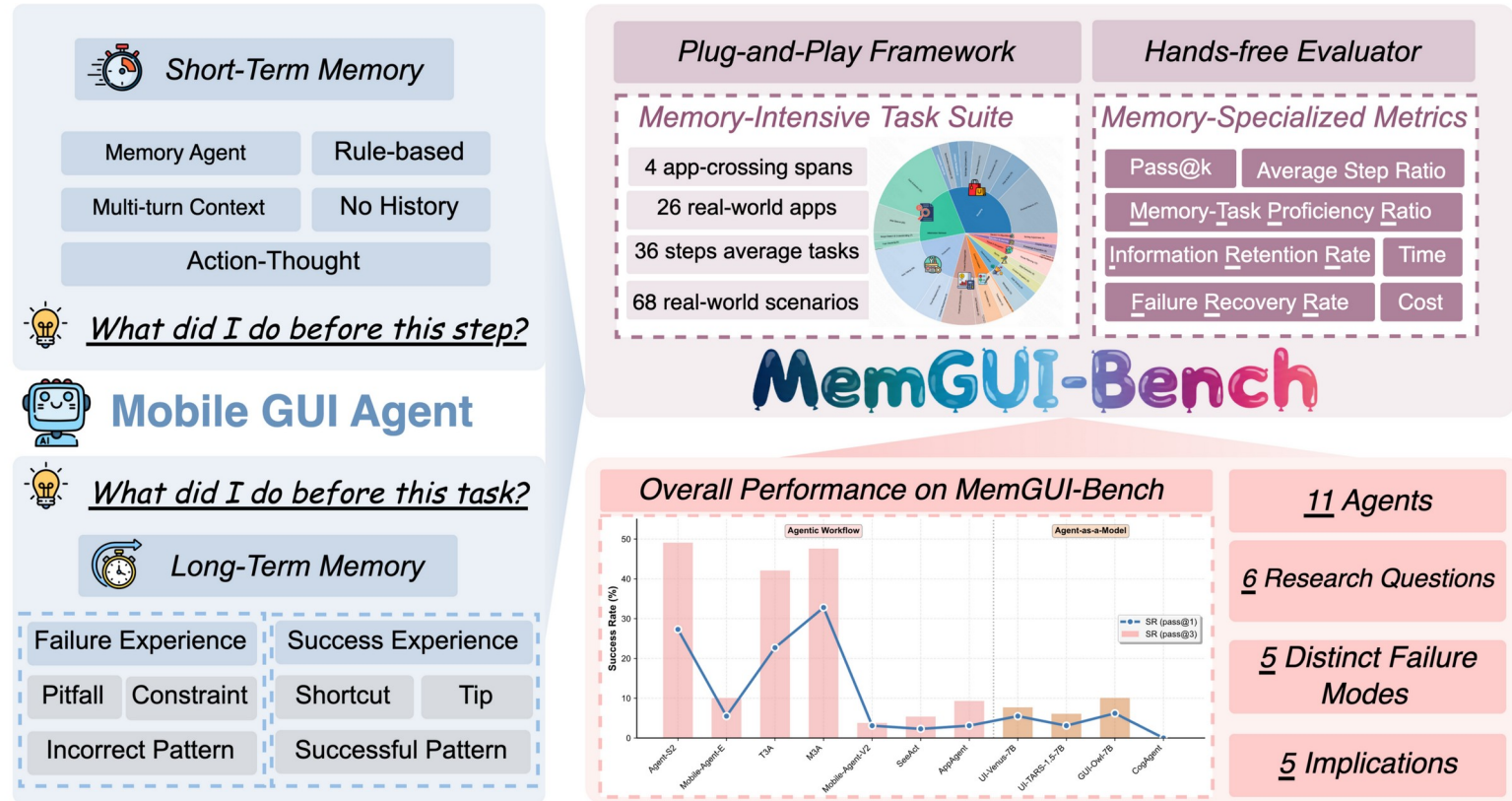
128 Tasks

26 Apps

11 Agents

arXiv:2602.06075

MemGUI-Bench | Taxonomy, Tasks, and Evaluation in One System



Evaluation Design

- Taxonomy: short-term retention and long-term cross-session learning
- Tasks: 128 tasks across 26 apps; 89.8% require cross-temporal or cross-spatial memory
- Protocol: pass@1 / pass@k for repeated attempts and learning
- MemGUI-Eval: three-stage Progressive Scrutiny and seven hierarchical metrics

A trustworthy benchmark is prerequisite to judging whether memory actually works

Findings | Standard Benchmarks Hide the Real Memory Gap

Six research questions across eleven state-of-the-art agents

4–10×

Memory capability gap

Standard vs memory-intensive tasks

–16–40 pp

Cross-app degradation

Primary memory bottleneck

+18.8 pp

Long-context gain

RQ4

+21.9 pp

Long-term memory gain

RQ5

FAILURE DIAGNOSIS Five failure modes: partial, process, and output memory hallucination, knowledge deficiency, and intent misunderstanding, leading to five design implications.

A trustworthy benchmark is prerequisite to judging whether memory actually works

MobileForge: Annotation-Free Adaptation for Mobile GUI Agents with Hierarchical Feedback-Guided Policy Optimization

Explore target apps, mine curricula, evaluate rollouts, and optimize policies without manual annotations

CORE QUESTION Target apps are numerous and fast-changing, making adaptation with human tasks, demonstrations, and reward labels unscalable.

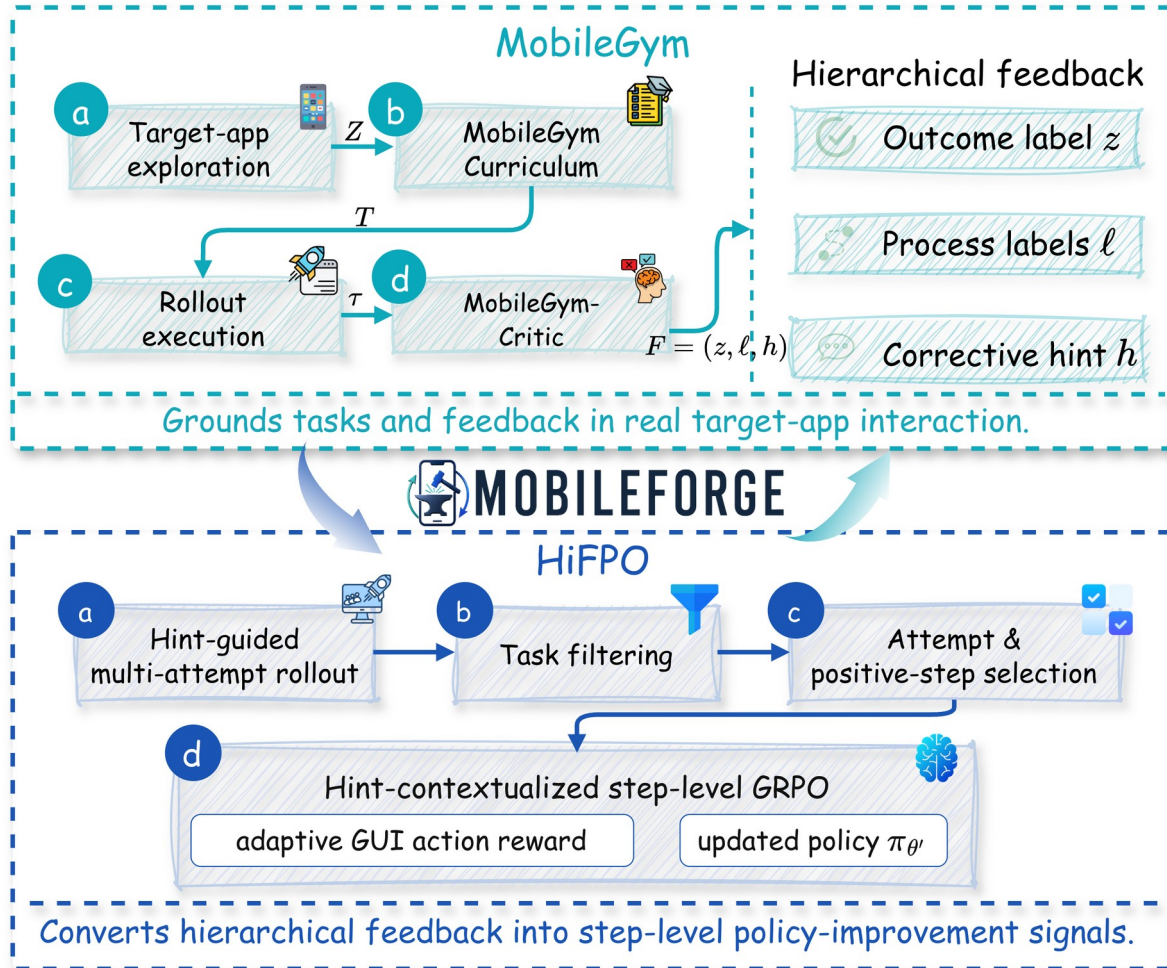
MobileGym

HiFPO

RL Post-Training

arXiv:2606.19930

MobileForge | A Real-Interaction Self-Improvement Loop



Two Coupled Components

- MobileGym: target-app exploration, curriculum mining, rollout execution, and hierarchical evaluation
- Hierarchical feedback: outcome labels, process labels, and corrective hints
- HiFPO: multi-attempt rollouts, task filtering, and positive-step selection
- Hint-contextualized step-level GRPO converts feedback into learnable signals

Turn real target apps into a continuously improving agent factory

Results | Automatically Generated Data Can Produce Strong Policies

Validated on AndroidWorld and MobileWorld using only automatically generated adaptation data

67.2%

Qwen3-VL-8B Pass@3

AndroidWorld; near the closed-data base at 69.0%

77.6%

ForgeOwl-8B Pass@3

AndroidWorld

41.0%

ForgeOwl-8B success

MobileWorld GUI-only

KEY RESULT

MobileForge improves both generalist and GUI-specialized models; ForgeOwl-8B is the strongest open-data mobile GUI agent in the paper.

Turn real target apps into a continuously improving agent factory

MemGUI-Agent: An End-to-End Long-Horizon Mobile GUI Agent with Proactive Context Management

Making context management a first-class action inside an end-to-end policy

CORE QUESTION ReAct passively accumulates history, causing prompt growth, noise, and dilution of critical cross-app facts on long-horizon tasks.

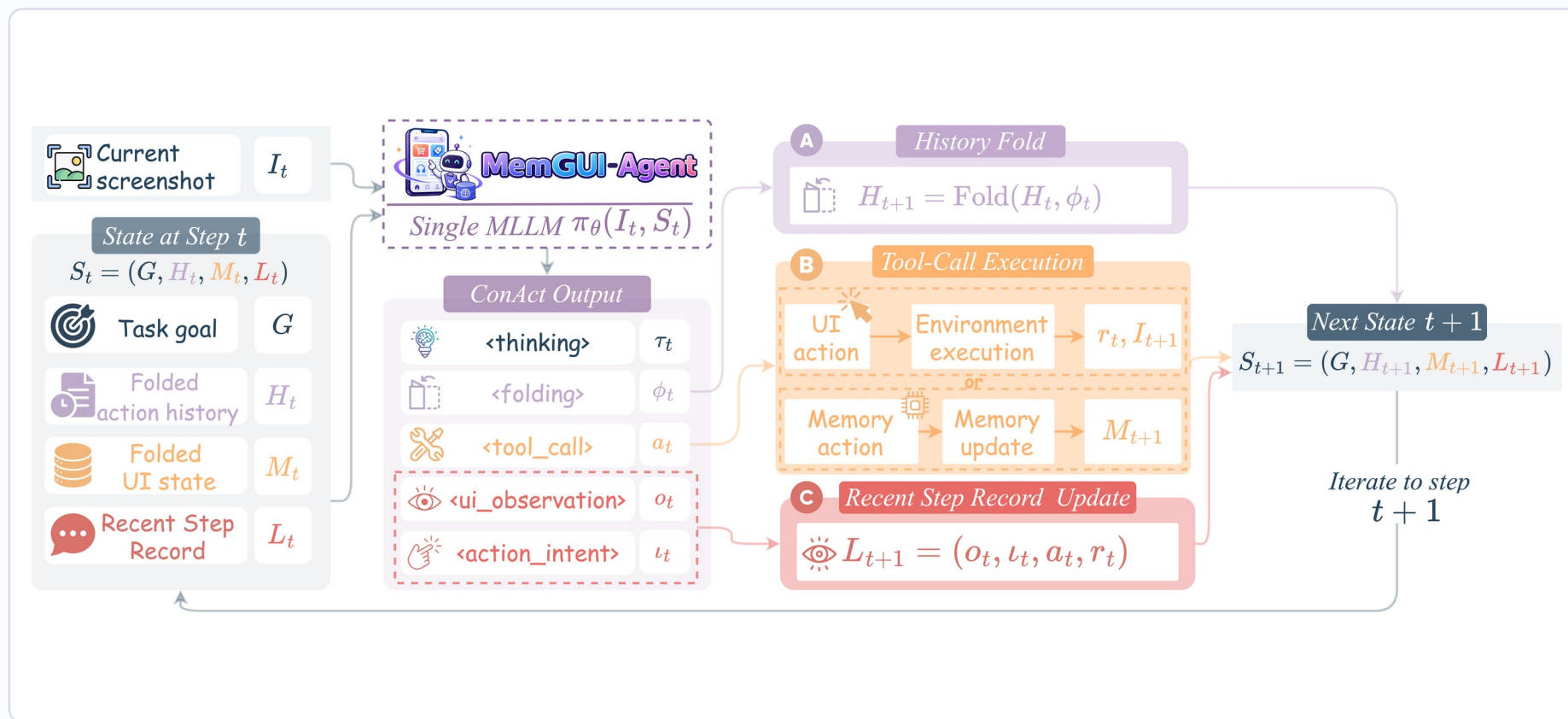
ConAct

MemGUI-3K

8B SFT

arXiv:2606.19926

ConAct | Context Management as Action



One Policy, Joint Decisions

- Folded action history compresses completed sub-tasks
- Folded UI state preserves critical interface facts
- Recent step record retains immediate observations and intent
- One forward pass emits reasoning, folding, tool call, observation, and next intent

A long-horizon agent must learn to manage its own working context while acting

Results | Lower Context Cost, Stronger Long-Horizon Performance

Zero-shot ConAct with 235B and supervised MemGUI-8B-SFT

~1.5k

Input tokens saved by step 150

Compared with ReAct

62.5%

235B MemGUI-Bench Pass@3

Pass@1 = 37.5%

2,956

MemGUI-3K trajectories

64,430 SFT samples

17.9%

8B MobileWorld success

+8.5 pp over backbone

TRANSFER

Zero-shot 235B ConAct sets a new MemGUI-Bench SOTA; training on MemGUI-3K gives the best open-data 8B result and transfers to MobileWorld.

A long-horizon agent must learn to manage its own working context while acting